

Project Proposal

CNN-LSTM/GRU Based Automatic Image Caption Generator

Course	Deep Learning (702AI0C008)
Domain	Computer Vision + Natural Language Generation (CNN – RNN Hybrid)
Student Name	_____
Roll No. / SAP ID	_____
Date	_____

1. Objective

To design and implement a hybrid deep learning system that automatically generates a natural-language description for a given input image, by combining a Convolutional Neural Network (CNN) for visual feature extraction with a Recurrent Neural Network (LSTM / GRU) for sequential language generation. The project additionally aims to conduct a controlled ablation study comparing LSTM and GRU decoders under identical conditions, to evaluate their relative performance for the image captioning task.

2. Motivation

Automatically describing the content of an image is a task that connects two major subfields of AI — computer vision and natural language processing. It has practical applications in accessibility tools (e.g., describing images to visually impaired users), image indexing and retrieval, and content moderation. This project was chosen because it is one of the few applications that meaningfully exercises both the convolutional architectures (Unit 4) and the recurrent / LSTM / GRU architectures (Unit 5) covered in this course within a single end-to-end pipeline, making it well suited to demonstrate applied understanding of the full course syllabus. It also remains an actively researched problem, with recent work (2023–2025) continuing to refine CNN-LSTM/GRU based approaches rather than abandoning them in favour of purely Transformer-based methods.

3. Literature Review (Mother Papers)

3.1 Paper 1: “An Image Captioning Algorithm Based on the Hybrid Deep Learning Technique (CNN+GRU)”

Reference: arXiv:2301.02440, published January 2023

Core Idea:

The paper proposes a CNN encoder – GRU decoder framework augmented with a novel **semantic reconstructor** module. The reconstructor uses the decoder's hidden states to reconstruct the semantic representation of the input image, and the resulting “reconstruction score” is combined with the standard likelihood score during both training and caption selection at inference time. This is intended to address a known weakness of plain CNN-RNN captioning models: performance degrades sharply on visually complex images due to weak semantic understanding of the scene.

Input → Output:

Input: a single image. Output: a caption sentence, selected using a combination of likelihood and semantic-reconstruction score.

Limitations:

- Adding the semantic reconstructor increases model complexity and training cost compared to a plain CNN-GRU baseline.
- The approach compensates for weak scene understanding after the fact (via reconstruction scoring) rather than allowing the model to dynamically attend to specific image regions while generating each word.
- Standard CNN-LSTM pairs are explicitly noted in the paper as suffering from poor time and space complexity efficiency, a trade-off this method partially addresses but does not eliminate.

3.2 Paper 2: “Image Captioning Deep Learning Model Using ResNet50 Encoder and Hybrid LSTM–GRU Decoder Optimized with Beam Search”

Authors: Kavitha P., Karpagam V. | **Venue:** Automatika, Vol. 66, No. 3, pp. 394–410, published online 16 April 2025

Core Idea:

The paper systematically compares several CNN encoders (VGG16, InceptionV3, ResNet50, DenseNet121) paired with an LSTM decoder on the Flickr8k dataset, then proposes a **hybrid LSTM–GRU decoder** combined with **beam search** decoding (instead of greedy search) as the best-performing configuration. The ResNet50 + Hybrid LSTM–GRU + Beam Search configuration achieves a reported accuracy of 0.8932, a loss of 0.4013, and a BLEU score of 0.6034 on Flickr8k, outperforming the greedy-search single-decoder baselines.

Input → Output:

Input: a single image. Output: a caption sentence generated via beam search over the hybrid LSTM–GRU decoder’s output distribution.

Limitations:

- Beam search adds inference-time computational overhead compared to greedy decoding.
- The hybrid decoder design increases architectural complexity without addressing the deeper issue common to both papers: the CNN-extracted image feature is used as a static input, with no attention mechanism to dynamically focus on relevant image regions per generated word.
- The systematic encoder comparison is thorough, but the paper does not isolate decoder type (LSTM vs. GRU) as a controlled standalone variable — it evaluates the hybrid combination directly rather than LSTM-only vs. GRU-only under identical conditions.

4. Proposed System / Methodology

The proposed system builds on the encoder-decoder framework used in both mother papers, and directly targets the specific gap left open by them: neither paper runs a controlled, isolated comparison of LSTM-only vs. GRU-only decoders, or of a frozen vs. fine-tuned CNN encoder, under otherwise identical conditions. It consists of the following stages:

4.1 Architecture

- **Encoder:** A pretrained CNN (ResNet-50, following Paper 2's best-performing encoder choice) extracts a fixed-length feature vector from each input image, used as a frozen feature extractor in the baseline configuration.
- **Bridge Layer:** A dense (fully connected) layer projects the CNN feature vector into the embedding dimension of the decoder, used to initialize its hidden state.
- **Decoder:** A recurrent network (LSTM in the primary model, GRU in the comparison model) combined with a word-embedding layer generates the caption one word at a time, conditioned on the image feature and the previously generated words.
- **Output Layer:** A softmax layer over the vocabulary predicts the next word at each timestep; the full caption is produced via greedy decoding and, as an extension inspired by Paper 2, beam search.

4.2 Dataset

Flickr8k (8,000 images, 5 human-annotated captions each) will be used as the primary dataset, consistent with both mother papers, given its manageable size for course-project-scale compute and training time. Flickr30k may be used as an extension if time and compute permit.

4.3 Novelty / Ablation Study — Addressing the Gap

Both mother papers improve on the base CNN-RNN captioning idea (via semantic reconstruction and via hybrid decoding with beam search, respectively), but neither isolates the individual contribution of decoder type or encoder training strategy through a controlled, side-by-side comparison. This project directly performs that missing comparison:

- **LSTM vs. GRU decoder** — identical encoder and training setup, decoder architecture varied, to empirically determine which recurrent unit performs better for this task and dataset in isolation.
- **Frozen vs. fine-tuned CNN encoder** — comparing a frozen pretrained ResNet50 encoder against a jointly fine-tuned encoder, to quantify the accuracy / compute trade-off directly.

Attention-based extensions (e.g., Show, Attend and Tell) remain outside this project's scope, as this aspect is common to both mother papers' limitations and would require architecture beyond the CNN/RNN/LSTM/GRU fundamentals covered in this course; it is explicitly noted as a direction for future work.

4.4 Evaluation Metrics

- BLEU-1 to BLEU-4 scores against reference captions.
- Qualitative evaluation on a held-out set of test images, including both successful and failure-case examples.
- Training time and convergence comparison across the ablation configurations.

5. System Requirements

Component	Specification
Compute (recommended)	Google Colab (free tier) – NVIDIA T4 GPU, 16 GB VRAM

Compute (local, optional)	CPU-only feature extraction is feasible on any modern i5/i7; GPU recommended only for decoding
Language / Framework	Python 3.9+, TensorFlow / Keras (or PyTorch)
Pretrained Models	ResNet-50 (via <code>keras.applications</code> or <code>torchvision.models</code>)
Supporting Libraries	NumPy, Pandas, NLTK (BLEU scoring), Matplotlib, tqdm
Storage	~1 GB (Flickr8k dataset) + checkpoints, stored on Google Drive
Dataset Source	Flickr8k (Kaggle / official release)

6. Expected Outcomes

- A working CNN–LSTM/GRU model capable of generating reasonably coherent captions for unseen images.
- Quantitative comparison (BLEU scores) between LSTM and GRU decoder variants, and between frozen and fine-tuned CNN encoders.
- A discussion of observed limitations (e.g., generic captions on complex scenes) in relation to the limitations identified in both mother papers.
- A clear articulation of how attention-based approaches could extend this work beyond the taught syllabus, positioned as future work.

7. Tentative Timeline

Week	Activity
1	Literature review of mother papers; dataset acquisition (Flickr8k)
2	Data preprocessing; CNN feature extraction pipeline
3	Baseline LSTM decoder implementation and training
4	GRU decoder implementation; frozen vs. fine-tuned CNN ablation
5	Evaluation (BLEU scoring), qualitative analysis, error analysis
6	Report writing, results compilation, and final presentation prep

8. References

- [1] M. A. Azhar et al., “An Image Captioning Algorithm Based on the Hybrid Deep Learning Technique (CNN+GRU),” *arXiv preprint*, arXiv:2301.02440, Jan. 2023.
- [2] P. Kavitha and V. Karpagam, “Image Captioning Deep Learning Model Using ResNet50 Encoder and Hybrid LSTM–GRU Decoder Optimized with Beam Search,” *Automatika*, vol. 66, no. 3, pp. 394–410, 2025. DOI: 10.1080/00051144.2025.2485695
- [3] S. Hochreiter and J. Schmidhuber, “Long Short-Term Memory,” *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [4] K. He, X. Zhang, S. Ren, and J. Sun, “Deep Residual Learning for Image Recognition,” in *Proc. IEEE CVPR*, 2016, pp. 770–778.

- [5] M. Hodosh, P. Young, and J. Hockenmaier, "Framing Image Description as a Ranking Task: Data, Models and Evaluation Metrics," *Journal of Artificial Intelligence Research*, vol. 47, pp. 853–899, 2013. (Flickr8k dataset)
- [6] O. Vinyals, A. Toshev, S. Bengio, and D. Erhan, "Show and Tell: A Neural Image Caption Generator," in *Proc. IEEE CVPR*, 2015, pp. 3156–3164. (Foundational reference – original CNN+LSTM captioning framework which papers [1] and [2] build upon)

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